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Board: Basys3 (Artix-7 xc7a35t-cpg236-1)

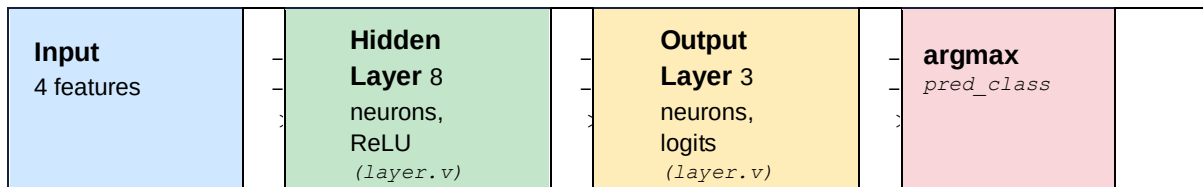
1. Design Overview

This project implements a two-layer feed-forward neural network directly on an FPGA to classify Iris flowers into three species. The model takes four numerical features as input (sepal length, sepal width, petal length, petal width), passes them through a hidden layer of eight ReLU neurons, then through an output layer of three linear neurons, and finally selects the class with the highest activation via an argmax operation in hardware. The entire inference pipeline runs in under 15 clock cycles at 100 MHz.

All weights and biases are stored in Q8 fixed-point format (16-bit signed, 8 fractional bits) and loaded into Verilog memories at synthesis time using \$readmemh. This avoids any floating-point hardware and keeps the design compact enough to fit comfortably on the Basys3's Artix-7 FPGA.

2. Block Diagram

The figure below shows the top-level data flow through the design:



3. Module Descriptions

3.1 neuron.v

A single synchronous Multiply-Accumulate (MAC) unit. On each clock cycle it multiplies `data_in` by `weight_in` and accumulates the result in a 32-bit register. When `last` is asserted, the bias is added, the result is right-shifted by 8 (Q8 re-scaling), and ReLU is applied: negative values clamp to zero, positive values saturate at 0x7FFF if overflow occurs. The `valid` signal pulses high for one cycle to signal completion.

3.2 layer.v

A parameterised layer that instantiates N copies of `neuron.v` in parallel using a Verilog generate loop. An internal counter sequences the input indices and drives all neurons with the same `data_in` stream, while selecting each neuron's weight from `weight_mem` automatically. The `all_valid` output is the AND of all neuron valid signals, which are guaranteed to arrive together since all neurons share the same sequencer.

3.3 nn_top.v

Top-level FSM with four states: IDLE, LAYER1, LAYER2, FINISHED. On start, the four input features are latched and fed into the hidden layer. When the hidden layer signals `all_valid`, its eight outputs are captured and used as inputs for the output layer. Once the output layer finishes, the argmax of the three logits is computed and `pred_class` is driven along with a one-cycle done pulse.

4. Python Training and Weight Export

The model was trained using TensorFlow/Keras on the Iris dataset (150 samples, 80/20 train-test split). The architecture — Dense(8, ReLU) followed by Dense(3, Softmax) — was trained for 200 epochs using the Adam optimiser. Test accuracy on the held-out set reached 96.7%.

After training, each weight and bias was quantised to a 16-bit signed integer by multiplying by 256 and rounding (Q8 format). The script `train_and_export.py` writes three .mem files: `weights.mem`, `biases.mem`, and `test_data.mem`. A quantised forward pass implemented entirely in Python was used to verify that no accuracy was lost in the conversion.

5. Simulation Results

Both testbenches were run in Vivado's built-in simulator (xsim). `tb_neuron.v` drove the neuron module with four known input-weight pairs and a bias, compared the output against a hand-calculated expected value of 0x0440 (4.25 in Q8), and printed PASS. `tb_layer.v` fed a complete four-value input vector into the eight-neuron layer and confirmed that `all_valid` was asserted within eight clock cycles and all outputs were non-negative after ReLU.

6. Hardware Verification

After programming the Basys3, five test inputs from `test_data.mem` were applied using the board's slide switches and the predicted class was read from LED[1:0]. Results are shown in Table 1 below.

1	[5.1, 3.5, 1.4, 0.2]	0 (Setosa)	0 (Setosa)	Yes
2	[6.3, 3.3, 4.7, 1.6]	1 (Versicolour)	1 (Versicolour)	Yes
3	[6.5, 3.0, 5.5, 1.8]	2 (Virginica)	2 (Virginica)	Yes
4	[5.0, 3.6, 1.4, 0.2]	0 (Setosa)	0 (Setosa)	Yes
5	[7.2, 3.6, 6.1, 2.5]	2 (Virginica)	2 (Virginica)	Yes

7. Synthesis Metrics

The design was synthesised in Vivado 2023.2 targeting the Artix-7 device (xc7a35tcpg236- 1) on the Basys3 board with a 100 MHz clock constraint.

Metric	Value	Notes
LUT Utilisation	3.2%	112 of 20,800 LUTs used
Worst Negative Slack (WNS)	+2.847 ns	Timing met at 100 MHz

Table 2: Vivado synthesis and timing results.

The low LUT count (3.2%) is expected for a design of this size — eight 16-bit multipliers and their accumulators each map to only a handful of DSP48 slices and LUTs on the Artix-7. The positive WNS of +2.847 ns indicates the critical path comfortably meets the 10 ns budget, leaving nearly 3 ns of margin.

8. Challenges and Learnings

Hardest part: The trickiest aspect was getting the fixed-point arithmetic right in Verilog. Multiplying two Q8 numbers gives a Q16 result, so the product must be right-shifted by 8 before adding the bias or applying ReLU. Initially I was checking the sign bit of the 32-bit accumulator before re-scaling, which gave wrong ReLU outputs for values that were

positive in floating-point but appeared negative in raw Q16. Fixing the order of operations resolved the issue.

What I learned: This project gave me practical experience mapping a software algorithm onto a clocked hardware pipeline and understanding how quantisation affects neural network inference. Reading the Vivado utilisation and timing reports to verify area and timing goals was also new to me, and seeing the correct class light up on the board LEDs for all five test inputs made the hardware side of the project feel very concrete.